

Bulletproofs

Short Proofs for Confidential Transactions and More

A Revolution of Zero-Knowledge Proofs in Blockchain

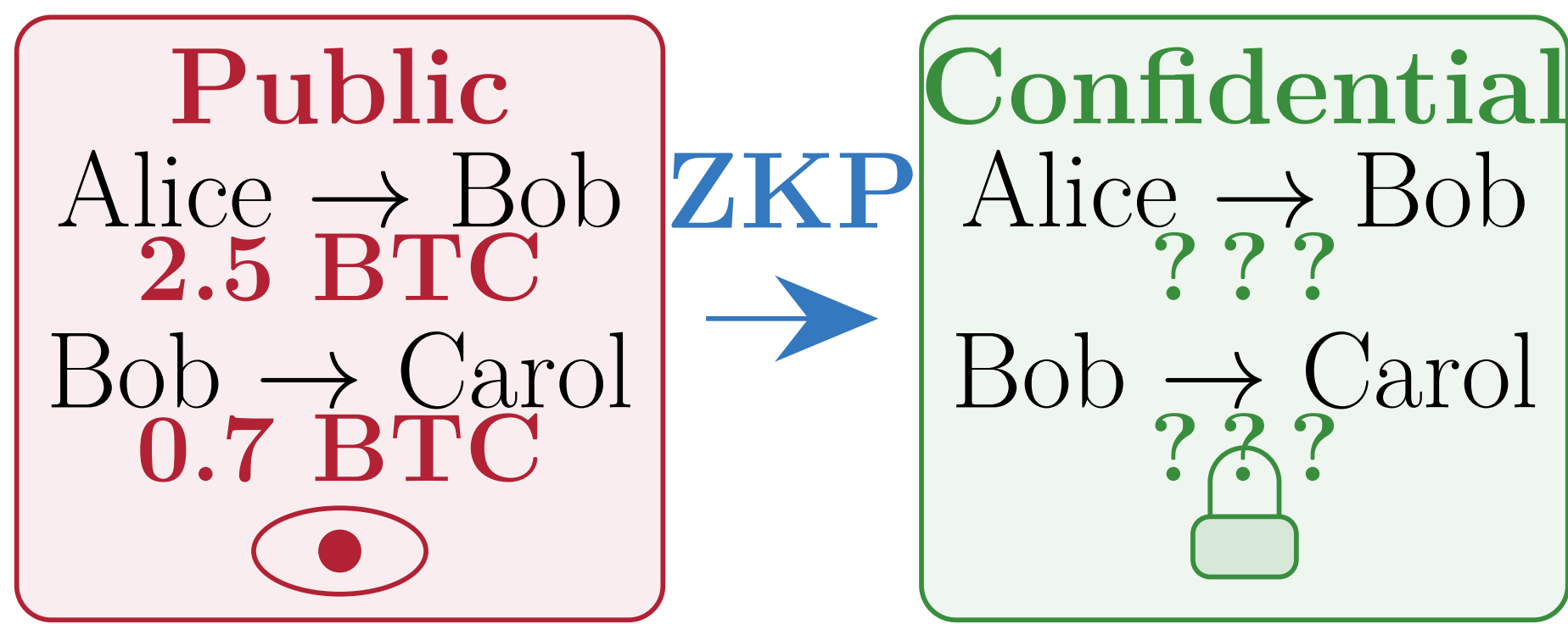
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The Big Idea

Bulletproofs let you prove a committed value lies in a range **without revealing it** — a proof under **1 KB**, **no trusted setup**.

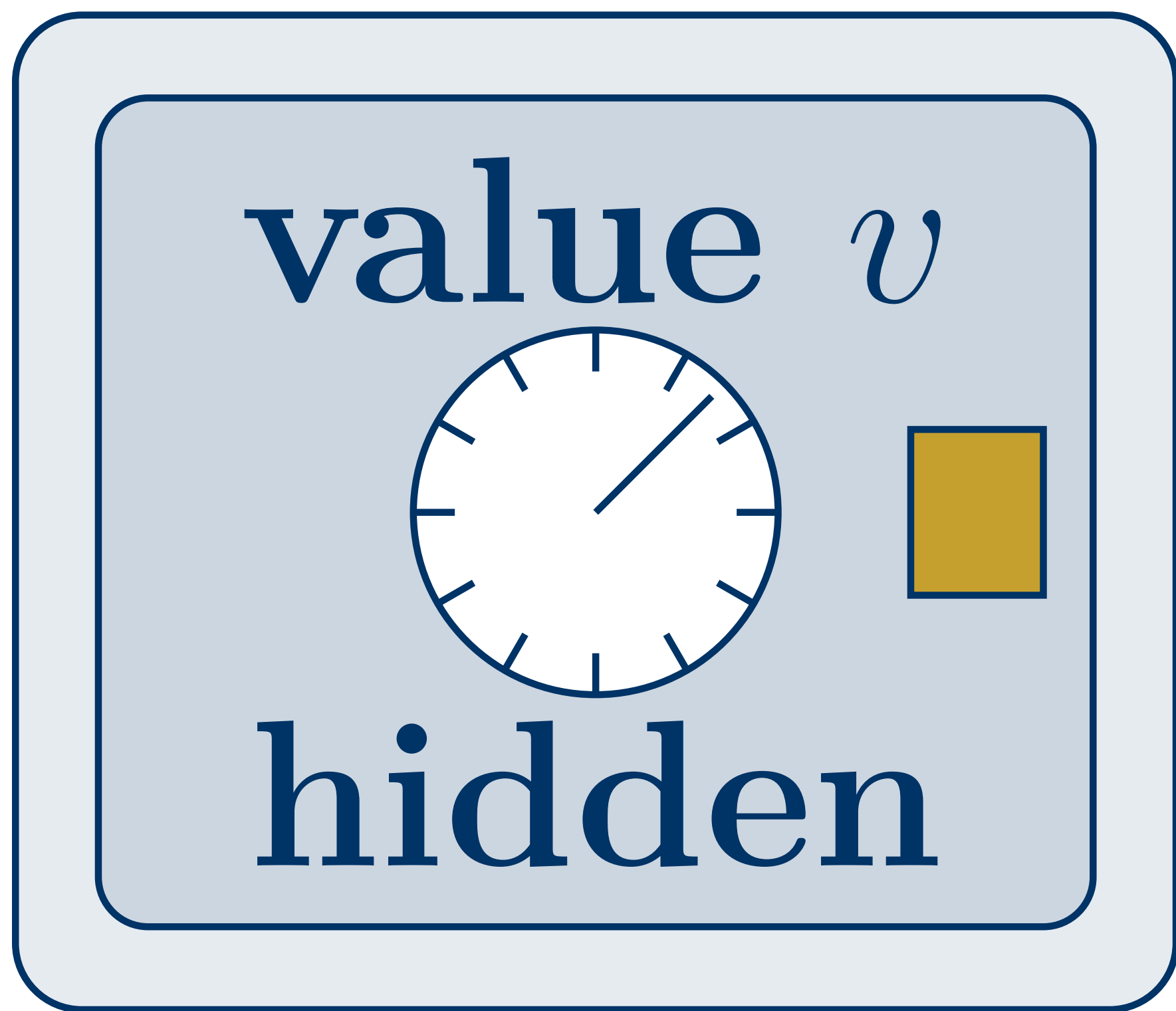
1. The Privacy Problem



“How can we prove an amount is non-negative without revealing it?”

2. Pedersen Commitment

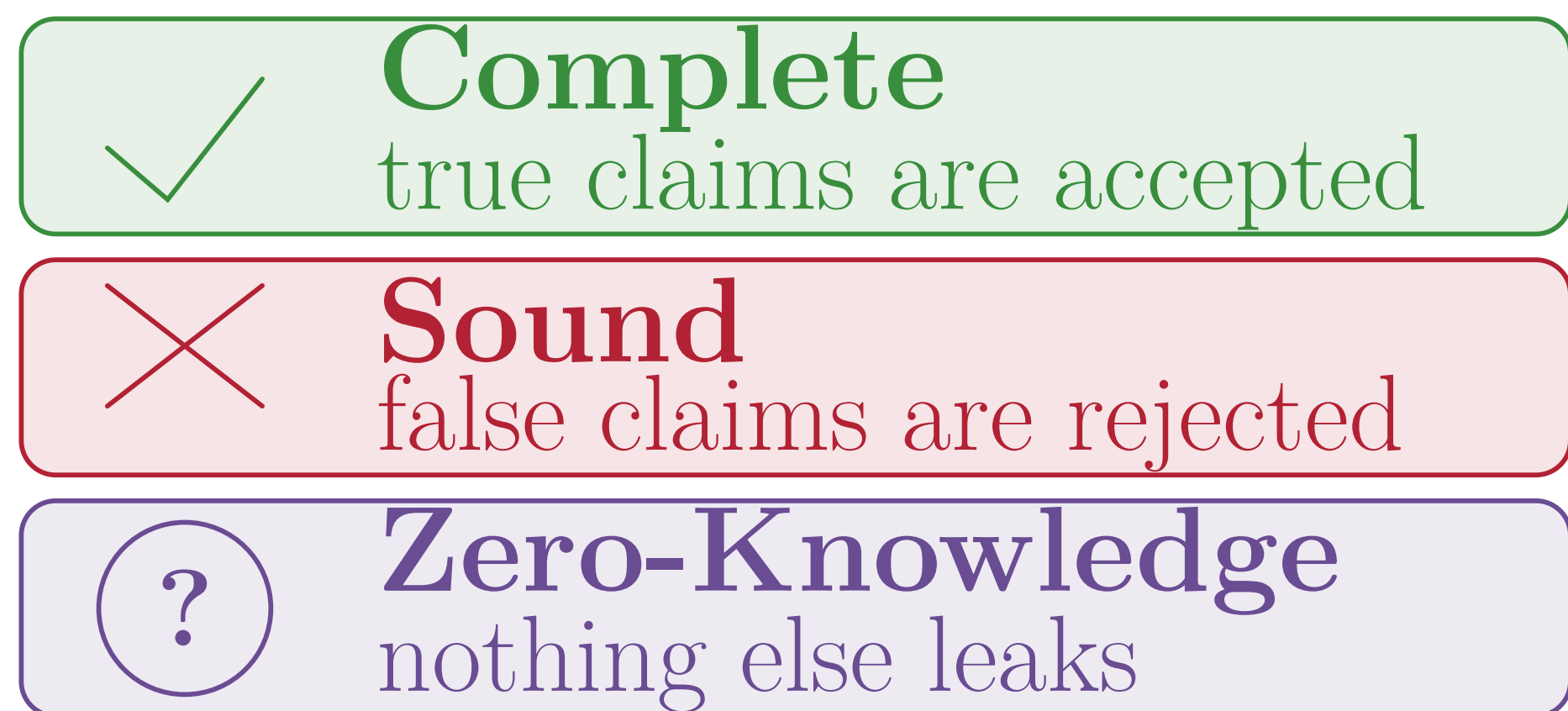
A commitment $V = g^v h^\gamma$ is like a **tamper-proof safe**: hides the value, but locks it in place.



- **Hiding** nobody learns v
- **Binding** v cannot change

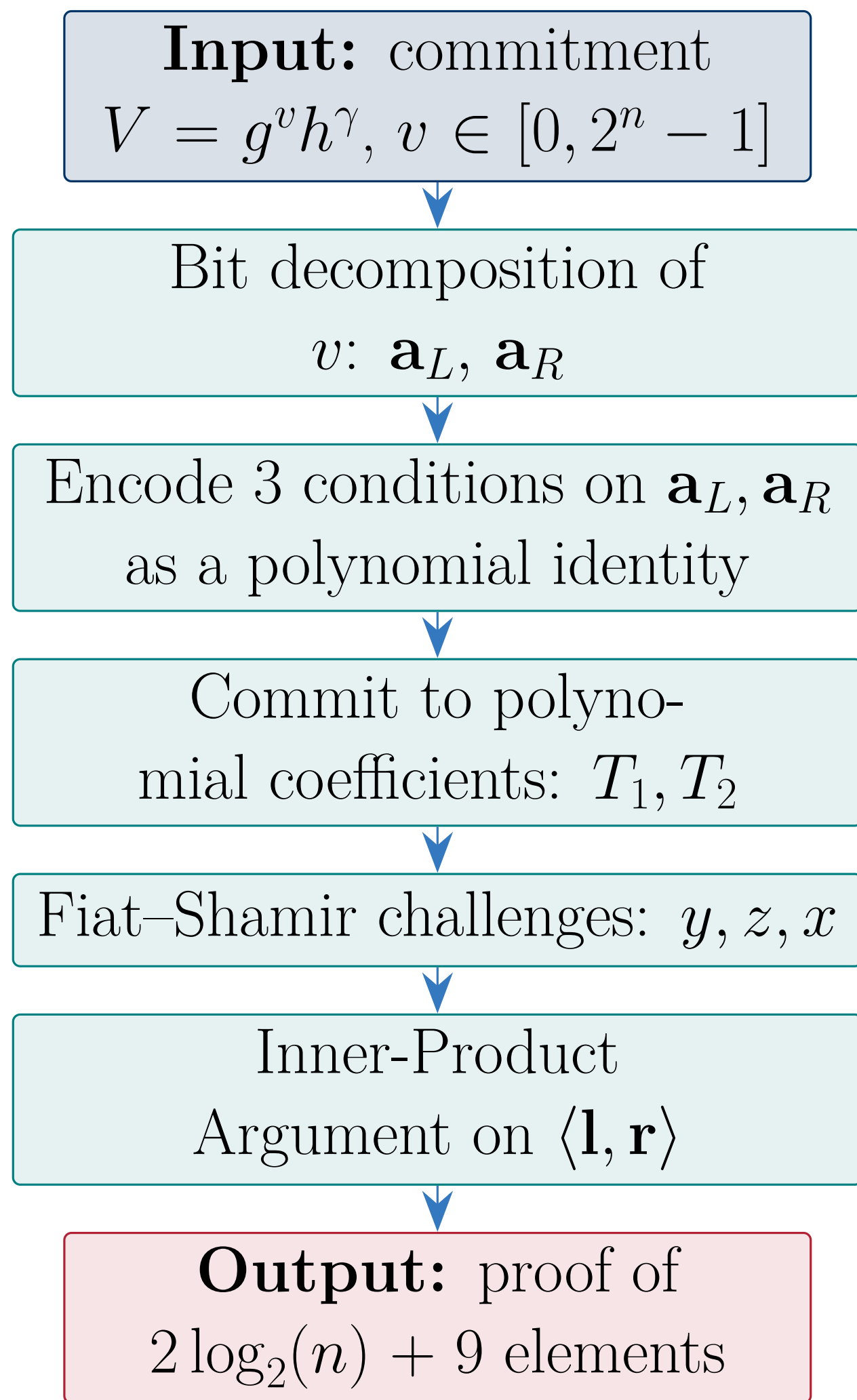
Magic: additively homomorphic
 $\text{Com}(v_1, \gamma_1) \cdot \text{Com}(v_2, \gamma_1) = \text{Com}(v_1 + v_2, \gamma_1 + \gamma_2)$. Arithmetic on locked values!

3. Three Pillars of ZKP



4. The Range Proof Flow

The diagram below summarises how Bulletproofs turn a range claim into a succinct logarithmic proof.



5. Main Result

Theorem (Bünz et al., 2018). For an n -bit range proof, Bulletproofs achieve — under the **discrete logarithm assumption** only and **without a trusted setup**:

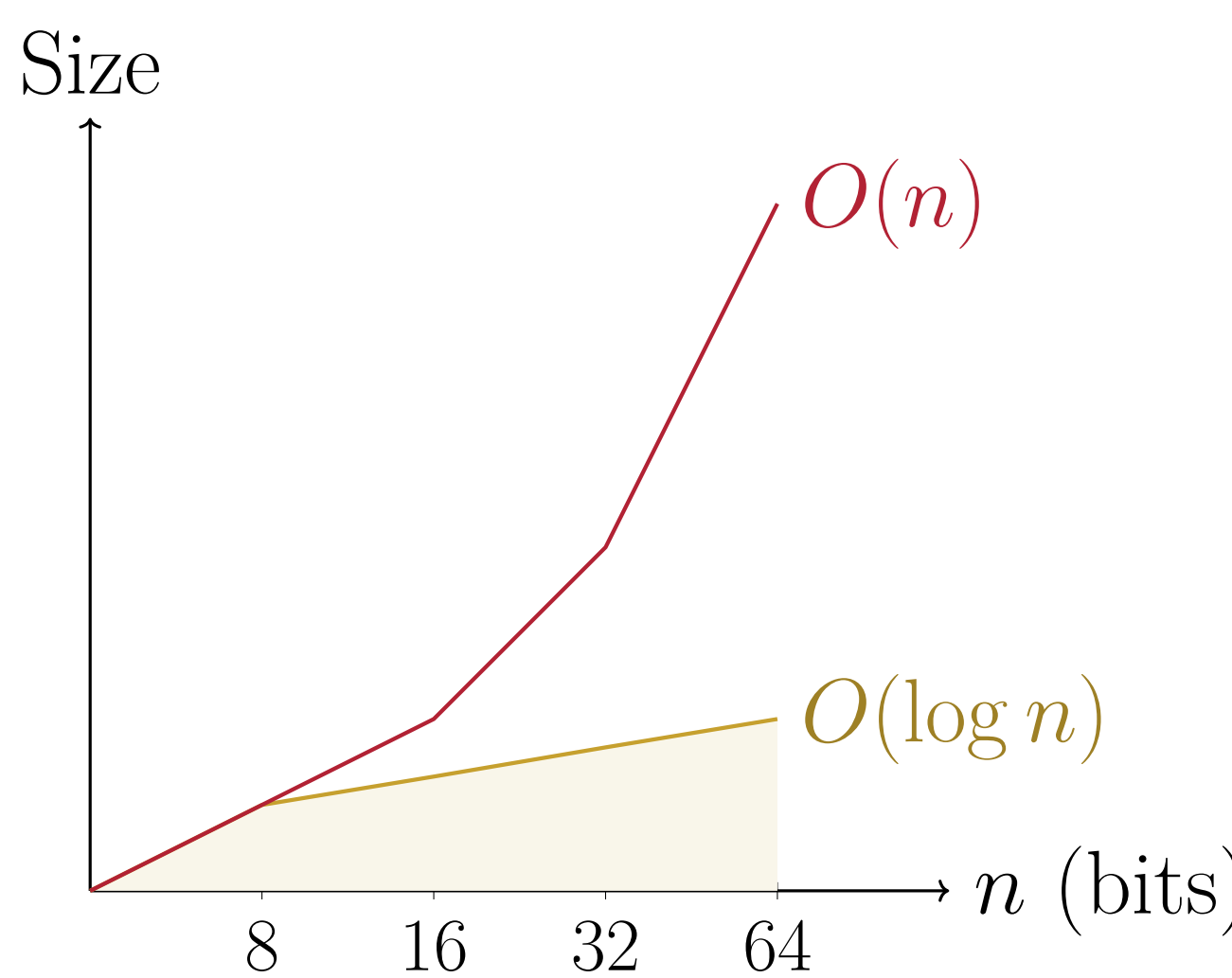
- **Proof size:** $2 \log_2(n) + 9$ group/field elements.
- **Prover time:** $O(n)$ group exponentiations.
- **Verifier time:** $O(n)$, amortised $O(n/\log n)$ in batch verification.
- **Aggregation:** m range proofs combine into a single proof of $2 \log_2(nm) + 9$ elements.

6. Performance Comparison

Proof sizes for a 64-bit range proof:

Protocol	Size	Trusted Setup
Borromean (Bitcoin CT)	~ 5 KB	No
zk-SNARK (Groth16)	~ 0,2 KB	Yes
Bulletproofs	~ 0,7 KB	No
zk-STARK	~ 100 KB	No

Proof size growth with bit length:



7. Applications

Bulletproofs are widely used across blockchain and privacy technologies:

- **Confidential Transactions:** hiding transaction amounts in Monero (CryptoNote), Grin and Beam (Mimblewimble).
- **Aggregated range proofs:** verifying many outputs together with a single short proof.
- **Solvency proofs:** exchanges proving they hold sufficient reserves without revealing wallet contents.
- **Arithmetic-circuit proofs:** general statements for smart contracts and complex predicates.
- **Privacy-preserving smart contracts:** confidential voting, sealed-bid auctions and private elections on Ethereum.

Real-world impact: In October 2018, Monero adopted Bulletproofs and reduced average transaction sizes by roughly **80%**, dramatically lowering fees. A clean example of an academic protocol scaling to a system with millions of users.

8. Conclusion

Bulletproofs strike a unique balance among zero-knowledge proof systems: minimal **size**, **no trusted setup**, and reliance only on the **discrete logarithm assumption**. While linear verification time remains their main drawback, follow-up work (Halo, Dory) builds directly on these ideas and underpins modern *recursive proof systems*.

Take-away: A meeting of mathematical elegance and cryptographic strength — inner-product arguments, vector polynomials and Pedersen commitments come together to shape the future of digital privacy.

References

1. B. Bünz, J. Bootle, D. Boneh, A. Poelstra, P. Wuille, G. Maxwell, *Bulletproofs: Short Proofs for Confidential Transactions and More*, IEEE Symposium on Security and Privacy, 2018, pp. 315–334.
2. J. Bootle, A. Cerulli, P. Chaidos, J. Groth, C. Petit, *Efficient Zero-Knowledge Arguments for Arithmetic Circuits in the Discrete Log Setting*, EUROCRYPT 2016.
3. G. Maxwell, *Confidential Transactions* (informal write-up), 2015. https://people.xiph.org/~greg/confidential_values.txt
4. T. P. Pedersen, *Non-Interactive and Information-Theoretic Secure Verifiable Secret Sharing*, CRYPTO 1991, pp. 129–140.
5. S. Bowe, J. Grigg, D. Hopwood, *Halo: Recursive Proof Composition without a Trusted Setup*, IACR ePrint Archive 2019/1021, 2019.

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